CIVITAS
Sustainable and smart mobility for all
2030

# CIVITAS indicators

## Citizens' willingness to replace polluting vehicles (ENV_PL_WL)

### DOMAIN

| Transport | **Environment** | Energy | Society | Economy |
|---|---|---|---|---|

| | | |
|---|---|---|
| **TOPIC** | **Pollution** | |
| **IMPACT** | **Composition of car fleet**<br>*Increasing citizens' willingness to replace polluting vehicles* | **ENV_PL** |

### Category

| Key indicator | **Supplementary indicator** | State indicator |
|---|---|---|

## CONTEXT AND RELEVANCE

Transport activity is a major source of urban pollution, affecting air quality, public health, and the climate. High-emission vehicles significantly contribute to greenhouse gas emissions and local air pollutants. Reducing the presence of internal combustion engine (ICE) vehicles in urban areas can therefore mitigate these impacts, supporting cleaner air and improved public health.

This indicator estimates the willingness of residents and users in the pilot area to replace their ICE car with an electric vehicle (EV). **It is a relevant indicator when the policy action is aimed at promoting EV uptake. A successful action is reflected in a <u>HIGHER</u> value of the indicator.**

## DESCRIPTION

The indicator is a **dimensionless** score summarising the willingness of residents and users of the pilot area to replace their ICE car with an electric car.

## METHOD OF CALCULATION AND INPUTS

The indicator is calculated as the average score provided by citizens in a sample survey. Respondents are asked to rate their willingness to replace their current ICE vehicle with an EV. **The indicator is computed exogenously**, following the method described below and implemented in the supporting tool.

| Method | |
|---|---|
| **Based on responses collected by sample survey** | Significance: **0.25** |

**The following information is needed** to compute the indicator:

a) **Responses of a sample of individuals to a survey question regarding their willingness to replace their ICE car with an electric car.**

A suggested formulation of the survey question on this topic is provided in the Guidelines for surveys which are part of the CIVITAS Evaluation Framework and is reported below:

Q: *How willing would you be to replace your current car with an electric vehicle? Please rate on a scale from 1 (not at all willing) to 10 (completely willing).*

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ | ○ |

The experiment would be reflected in terms of a different responses to the same question.

METHOD OF CALCULATION

The requirement for computing the indicator is collecting the responses from a sample survey (which can be organised to collect more information than the one needed for this indicator).

Assuming that the formulation of the question suggested in the Guidelines for surveys is used, the indicator should be computed according to the following steps, then entered in the supporting tool:

- **Calculation of the share of individuals assigning a certain score to their willingness to replace their ICE car with an EV** (see equation below)
- **Calculation of the indicator** (see equation below).

## EQUATIONS

The share of individuals assigning a certain score to their willingness to replace their ICE car with an EV should be computed as:

$$^{l}EVScoreSh = \frac{\sum_i {}^{f}EVScore_i \; where \; f = l}{I}$$

Where:

$^{f}EVScore_i$ = Score *f* assigned by individual *i*.

$I$ = Total number of responses collected.

For example, if a sample of 300 individuals was surveyed and 28 of them assigned an score of 7 to their willingness to adopt an EV, $\sum_i {}^{f}EVScore_i \; where \; f = 7$ would be 28 and the share of those assigning this score would be 28 / 300 = 9%

**The indicator should then be computed as:**

$$\boldsymbol{AvEVScore = \sum_{l} \left( {}^{l}EVScoreSh * {}^{l}EVScore \right)}$$

Where:

$^{l}EVScore$ = score *l* (i.e., the values from 1 to 10)